

## Practice Exam 1

MTH 207 (Fall 2026) — J. Baggett

| First Name | Last Name |
|------------|-----------|
|            |           |

Before exam begins:

- Write your first name and last name above.
- Turn off all electronics and keep them out of sight: no cellular phones, not even to tell time (and not even if it's just in airplane mode). No iPods, wearing of headphones, etc.

As soon as exam starts:

- Take a quick breath to relax!!
- Check that you have all 9 pages of the exam. (The number of pages includes this cover page.)

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK:**
  - Explain your reasons and justify your steps.
  - Demonstrate that you understand the *process*.
  - Present your logic in an organized manner on paper.
- Keep your eyes on your own exam!
- No calculators, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.

The real exam is 55 minutes and 100 points. This practice exam is deliberately longer (120 points) so it covers more of the material — do not use it to judge timing.

- When you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

(3<sup>pts</sup> ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

- (a) If the average rate of change of  $f$  on  $[1, 3]$  equals the average rate of change on  $[3, 5]$ , then  $f$  is linear on  $[1, 5]$ .

**Solution:**

**FALSE.** Only the three endpoint values are constrained.  $f$  can wander anywhere in between and still hit  $f(1)$ ,  $f(3)$ ,  $f(5)$  in arithmetic progression.

- (b) If  $\lim_{x \rightarrow 5} f(x) = 2$  and  $\lim_{x \rightarrow 5} g(x) = 0$ , then  $\lim_{x \rightarrow 5} \frac{f(x)}{g(x)}$  does not exist.

**Solution:**

**TRUE.** A nonzero numerator over a vanishing denominator grows without bound.

- (c) The graph of  $y = \frac{3x}{\sqrt{x^2+1}}$  approaches the same horizontal line as  $x \rightarrow \infty$  and as  $x \rightarrow -\infty$ .

**Solution:**

**FALSE.** Dividing inside the root means dividing by  $\sqrt{x^2} = |x|$ , which is  $x$  on the right and  $-x$  on the left. So the limit is 3 as  $x \rightarrow \infty$  and  $-3$  as  $x \rightarrow -\infty$  — two different horizontal asymptotes.

- (d) If  $s(t)$  is position, then  $s'(a)$  is the instantaneous velocity at time  $a$ , and it can be negative.

**Solution:**

**TRUE.** The derivative carries a sign; negative velocity means motion in the negative direction.

- (8<sup>pts</sup>) **2.** Let  $f(x) = x^2 - 3x$ . Find the slope of the secant line through the points  $(1, f(1))$  and  $(1+h, f(1+h))$ , and simplify it. What value does that slope approach as  $h \rightarrow 0$ , and what is that value called?

**Solution:**

$f(1) = 1 - 3 = -2$  and  $f(1+h) = (1+h)^2 - 3(1+h) = h^2 - h - 2$ , so

$$m_{\text{sec}} = \frac{f(1+h) - f(1)}{(1+h) - 1} = \frac{(h^2 - h - 2) - (-2)}{h} = \frac{h^2 - h}{h} = \frac{h(h-1)}{h} = h - 1$$

(the cancellation is legal because  $h \neq 0$ ). As  $h \rightarrow 0$  this approaches  $-1$ , which is  $f'(1)$  — the **derivative** of  $f$  at 1, i.e. the slope of the tangent line at  $(1, -2)$ .

(18<sup>pts</sup>) **3.** Evaluate the following limits. As a reminder, **justify your steps!** Simplify your answer as much as possible. (If the limit does not exist, state why.)

- $\lim_{x \rightarrow -1} (2x^2 - 5x + 4)$

- $\lim_{t \rightarrow 3} \frac{t^2 + t - 12}{t - 3}$

- $\lim_{x \rightarrow 0} x^4 \cos\left(\frac{1}{x}\right)$

**Solution:**

(a) A polynomial is continuous everywhere, so substitute directly:  $2(-1)^2 - 5(-1) + 4 = 2 + 5 + 4 = \mathbf{11}$ .

(b) Substituting gives  $\frac{0}{0}$ , so factor first:

$$\lim_{t \rightarrow 3} \frac{t^2 + t - 12}{t - 3} = \lim_{t \rightarrow 3} \frac{(t + 4)(t - 3)}{t - 3} = \lim_{t \rightarrow 3} (t + 4) = \mathbf{7}$$

Cancelling  $t - 3$  is legal because  $t \neq 3$  inside the limit.

(c)  $\cos\left(\frac{1}{x}\right)$  has no limit at 0, but it is bounded:  $-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$ . Multiplying by  $x^4 \geq 0$ ,

$$-x^4 \leq x^4 \cos\left(\frac{1}{x}\right) \leq x^4$$

Both bounds tend to 0, so by the **Squeeze Theorem** the limit is **0**.

(18pts) 4. Evaluate the following limits. As a reminder, **justify your steps!** (If the limit does not exist, state why.)

- $\lim_{x \rightarrow 1} \frac{x-1}{|x-1|}$
- $\lim_{x \rightarrow 0} \frac{\cos x}{2x + \sin x}$
- $\lim_{x \rightarrow 2} e^{x^2-3x}$

**Solution:**

(a) For  $x > 1$ ,  $|x-1| = x-1$  and the quotient is 1. For  $x < 1$ ,  $|x-1| = -(x-1)$  and the quotient is  $-1$ . The one-sided limits are 1 and  $-1$ ; they disagree, so the limit **does not exist**.

(b) The numerator tends to  $\cos 0 = 1$ , and the denominator  $2x + \sin x$  tends to 0 — so this is *not*  $\frac{0}{0}$  and nothing cancels. Just right of 0 the denominator is small and *positive*, so the quotient runs to  $+\infty$ ; just left it is small and negative, so the quotient runs to  $-\infty$ . The two sides disagree, so the limit **does not exist**.

(c)  $e^x$  and  $x^2 - 3x$  are both continuous, so their composition is, and we substitute:  $e^{2^2-3(2)} = \mathbf{e^{-2}}$ .

(8pts) 5. Evaluate  $\lim_{x \rightarrow 0} \frac{\sqrt{x+4}-2}{x}$ . (Hint: rationalize the numerator.)

**Solution:**

Substituting gives  $\frac{0}{0}$ , so the quotient has to be rewritten. Multiply by the conjugate:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x+4}-2}{x} &= \lim_{x \rightarrow 0} \frac{(\sqrt{x+4}-2)(\sqrt{x+4}+2)}{x(\sqrt{x+4}+2)} = \lim_{x \rightarrow 0} \frac{(x+4)-4}{x(\sqrt{x+4}+2)} \\ &= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+4}+2)} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+4}+2} = \frac{1}{2+2} = \frac{1}{4} \end{aligned}$$

Keep lim on every line until the last: the cancellation  $\frac{x}{x} = 1$  is only legal because  $x \neq 0$ , and the limit symbol is what says so.

(7<sup>pts</sup>) **6.** State the Squeeze Theorem.

**Solution:**

If  $f(x) \leq g(x) \leq h(x)$  near  $a$  (except possibly at  $a$  itself), and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L,$$

then  $\lim_{x \rightarrow a} g(x) = L$ .

Three hypotheses, and the third is the one that gets dropped: the two outer limits must **agree**. Trapped is not enough. The point is that  $g$  never needs a formula — it only has to be caught between two functions that meet.

(10<sup>pts</sup>) **7.** Let  $f(x) = \frac{x+1}{x-3}$ . Find  $\lim_{x \rightarrow 3^+} f(x)$  and  $\lim_{x \rightarrow 3^-} f(x)$ , and say what this tells you about the graph of  $f$ .

**Solution:**

Near  $x = 3$  the numerator  $x + 1$  is close to 4, so it is *positive and not small*. The denominator  $x - 3$  is what is small.

**From the right** ( $x > 3$ ):  $x - 3$  is small and *positive*, so  $\lim_{x \rightarrow 3^+} f(x) = +\infty$ .

**From the left** ( $x < 3$ ):  $x - 3$  is small and *negative*, so  $\lim_{x \rightarrow 3^-} f(x) = -\infty$ .

Because at least one one-sided limit is infinite, the line  $\mathbf{x = 3}$  is a **vertical asymptote** of the graph. The two-sided limit does not exist.

- (11pts) 8. Find the horizontal asymptotes of  $f(x) = \frac{3x^2 + 8}{7x^2 + 5}$ .

**Solution:**

Divide the numerator and denominator by the highest power in the denominator,  $x^2$ :

$$\lim_{x \rightarrow \pm\infty} \frac{3x^2 + 8}{7x^2 + 5} = \lim_{x \rightarrow \pm\infty} \frac{3 + \frac{8}{x^2}}{7 + \frac{5}{x^2}} = \frac{3 + 0}{7 + 0} = \frac{3}{7}$$

The same value comes out on both sides, so there is one horizontal asymptote,  $y = \frac{3}{7}$ .  
(Equal degrees, so it is the ratio of the leading coefficients.)

- (8pts) 9. Evaluate the following limit or state that it does not exist. Show the steps!

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + 2x} - x)$$

**Solution:**

This is  $\infty - \infty$ , which is not a number and not something you may simplify term by term. Multiply by the conjugate:

$$\begin{aligned} \lim_{x \rightarrow \infty} (\sqrt{x^2 + 2x} - x) &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + 2x} - x)(\sqrt{x^2 + 2x} + x)}{\sqrt{x^2 + 2x} + x} = \lim_{x \rightarrow \infty} \frac{(x^2 + 2x) - x^2}{\sqrt{x^2 + 2x} + x} \\ &= \lim_{x \rightarrow \infty} \frac{2x}{\sqrt{x^2 + 2x} + x} \end{aligned}$$

Now divide top and bottom by  $x$ . Out at  $+\infty$  we have  $x > 0$ , so  $x = \sqrt{x^2}$  — which is how the  $x$  gets *inside* the root:

$$\begin{aligned} &= \lim_{x \rightarrow \infty} \frac{\frac{2x}{x}}{\frac{\sqrt{x^2 + 2x}}{x} + \frac{x}{x}} && \text{divide top and bottom by } x \\ &= \lim_{x \rightarrow \infty} \frac{2}{\frac{\sqrt{x^2 + 2x}}{\sqrt{x^2}} + 1} && \text{here: } x = \sqrt{x^2}, \text{ because } x > 0 \\ &= \lim_{x \rightarrow \infty} \frac{2}{\sqrt{\frac{x^2 + 2x}{x^2}} + 1} && \text{one root over another is the root of the quotient} \\ &= \lim_{x \rightarrow \infty} \frac{2}{\sqrt{1 + \frac{2}{x}} + 1} && \text{divide inside} \\ &= \frac{2}{\sqrt{1 + 0} + 1} = \frac{2}{1 + 1} = 1 \end{aligned}$$

Had this run to  $-\infty$  instead, the only change would be  $x = -\sqrt{x^2}$ , and a minus sign would ride through every line below it.

- (15<sup>pts</sup>) **10.** Let  $f(x) = 3 + \frac{4}{x}$ . Use the **definition of derivative** to compute  $f'(5)$ . Note that using the Power Rule  $\frac{d}{dx}(x^n) = nx^{n-1}$  will not earn points.

**Solution:**

$$f'(5) = \lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h} = \lim_{h \rightarrow 0} \frac{\left(3 + \frac{4}{5+h}\right) - \left(3 + \frac{4}{5}\right)}{h} = \lim_{h \rightarrow 0} \frac{\frac{4}{5+h} - \frac{4}{5}}{h}$$

Put the numerator over a common denominator:

$$\frac{4}{5+h} - \frac{4}{5} = \frac{4 \cdot 5 - 4(5+h)}{5(5+h)} = \frac{-4h}{5(5+h)}$$

The main bar is a division, so multiply by  $\frac{1}{h}$ :

$$f'(5) = \lim_{h \rightarrow 0} \frac{-4h}{5(5+h)} \cdot \frac{1}{h} = \lim_{h \rightarrow 0} \frac{-4}{5(5+h)} = \frac{-4}{5 \cdot 5} = -\frac{4}{25}$$

- (3<sup>pts</sup>) **11.** Show by means of an example that  $\lim_{x \rightarrow a} [f(x) + g(x)]$  may exist even though neither  $\lim_{x \rightarrow a} f(x)$  nor  $\lim_{x \rightarrow a} g(x)$  exists.

**Solution:**

Take  $a = 0$  with

$$f(x) = \frac{1}{x}, \quad g(x) = -\frac{1}{x}$$

Neither  $\lim_{x \rightarrow 0} f(x)$  nor  $\lim_{x \rightarrow 0} g(x)$  exists (each runs to  $+\infty$  on one side and  $-\infty$  on the other). But for every  $x \neq 0$ ,  $f(x) + g(x) = 0$ , so  $\lim_{x \rightarrow 0} [f(x) + g(x)] = 0$ , which certainly exists.

(The Sum Law says the limit of a sum *is* the sum of the limits **when both limits exist**. This shows why that hypothesis is not optional — it cannot be read backwards.)

- (4<sup>pts</sup>) **12.** Give a formula for a function that is continuous at  $x = 2$  but **not** differentiable at  $x = 2$ . Briefly say why your example works.

**Solution:**

$f(x) = |x - 2|$  works.

It is **continuous** at 2:  $\lim_{x \rightarrow 2} |x - 2| = 0 = f(2)$ .

It is **not differentiable** at 2, because the difference quotient has different one-sided limits:

$$\frac{f(2+h) - f(2)}{h} = \frac{|h|}{h} = \begin{cases} 1 & h > 0 \\ -1 & h < 0 \end{cases}$$

so  $\lim_{h \rightarrow 0} \frac{|h|}{h}$  does not exist. The graph has a **corner** at  $x = 2$ .

- (10<sup>pts</sup>) **13.** Find the  $x$ -coordinate(s) of any point(s) where the tangent line to the curve  $y = x^4 + 1$  is parallel to the line  $32x - y = 15$ .

**Solution:**

Rewrite the line as  $y = 32x - 15$ , so its slope is 32. Parallel means the tangent has the same slope, so solve  $y' = 32$ .

With  $y = x^4 + 1$ , the Power Rule gives  $y' = 4x^3$ , so

$$4x^3 = 32 \implies x^3 = 8 \implies \mathbf{x = 2}$$

$x^3 = 8$  has only the one real solution, so  $x = 2$  is the only such point. (Its  $y$ -value is  $2^4 + 1 = 17$ , if the point itself is wanted.)

**THIS IS THE LAST PAGE OF THE PRACTICE EXAM.**